Experiment No: 04

Title: Implementation of Different Types of Operators in SQL with Suitable Examples

Topics to be covered:

- Arithmetic Operators
- Logical Operators
- Comparison Operators
- Special Operators
- Set Operations

1. Arithmetic Operators

Arithmetic operators perform mathematical calculations on numeric data types.

Examples:

- + (Addition): Adds two numbers.
- - (Subtraction): Subtracts one number from another.
- * (Multiplication): Multiplies two numbers.
- / (Division): Divides one number by another.
- % (Modulo): Returns the remainder of a division operation.

Example Query:

```
SELECT salary, salary + 500 AS Increased_Salary,
       salary - 500 AS Decreased_Salary,
       salary * 2 AS Double_Salary,
       salary / 2 AS Half_Salary,
       salary % 3 AS Remainder
FROM employees;
```

2. Logical Operators

Logical operators are used to combine multiple conditions in SQL queries.

Examples:

- AND: Returns true if all conditions are true.
- OR: Returns true if at least one condition is true.
- NOT: Reverses the result of a condition.

Example Query:

```
SELECT * FROM employees
WHERE department = 'IT' AND salary > 60000;

SELECT * FROM employees
WHERE department = 'HR' OR department = 'Sales';

SELECT * FROM employees
WHERE NOT department = 'Marketing';
```

3. Comparison Operators

Comparison operators compare two values and return a boolean result.

Examples:

- = (Equal to)
- != or <> (Not equal to)
- > (Greater than)
- < (Less than)
- >= (Greater than or equal to)
- <= (Less than or equal to)

Example Query:

```
SELECT * FROM employees
WHERE salary > 70000;

SELECT * FROM employees
WHERE department <> 'IT';
```

4. Special Operators

Special operators are used for pattern matching and handling NULL values.

Examples:

- BETWEEN: Checks if a value is within a specified range.
- IN: Checks if a value exists within a given list.
- LIKE: Searches for a pattern in a column.
- IS NULL: Checks if a value is NULL.
- IS NOT NULL: Checks if a value is NOT NULL.

Example Query:

```
SELECT * FROM employees
WHERE salary BETWEEN 60000 AND 80000;

SELECT * FROM employees
WHERE department IN ('IT', 'HR', 'Sales');

SELECT * FROM employees
WHERE first_name LIKE 'J%';

SELECT * FROM employees
WHERE date_of_birth IS NULL;
```

5. Set Operations

Set operations combine the results of two or more queries.

Examples:

- UNION: Combines the results of two queries and removes duplicates.
- UNION ALL: Combines the results of two queries without removing duplicates.
- INTERSECT: Returns only common records between two queries.
- EXCEPT or MINUS: Returns records present in the first query but not in the second.

Example Query:

```
SELECT first_name FROM employees WHERE department = 'IT'
UNION
SELECT first_name FROM employees WHERE department = 'HR';
```

```
SELECT first_name FROM employees WHERE department = 'Sales'
INTERSECT
SELECT first_name FROM employees WHERE department =
'Marketing';

SELECT first_name FROM employees WHERE department = 'IT'
EXCEPT
SELECT first_name FROM employees WHERE department = 'HR';
```

<u>Lab Tasks-1:</u> Practicing SQL Arithmetic Operators

Example Table: Employees

employee_id	first_name	last_name	salary	hire_date	department	date_of_birth	gender	city
1	John	Doe	60000	2020-01-15	IT	1985-08-12	M	New York
2	Jane	Smith	70000	2019-03-23	HR	1990-05-22	F	Chicago
3	Tom	Brown	75000	2021-07-01	Sales	1988-12-04	M	Boston
4	Alice	Green	85000	2018-11-05	IT	1987-07-19	F	Miami
5	Bob	White	50000	2022-09-12	Marketing	1995-03-30	M	Dallas

1. Find the salary after adding a 10% bonus to each employee.
2. Retrieve employees who work in either the IT or HR department.
3. Display employees whose salary is greater than 70000.
4. Find employees whose first name starts with "A" or "J".
5. Retrieve all employees except those in the "Marketing" department.
6. Display employees whose salary falls between 50000 and 80000.
7. Find employees who have NULL values in their date_of_birth column.
8. List employees who are common between IT and HR departments using INTERSECT.
9. Show all employees from IT and HR without duplicates using UNION.
10. Retrieve employees from IT who are not in HR using EXCEPT.

<u>Lab Task-2:</u> Practicing SQL Arithmetic Operators
Example Table: Employees

```
CREATE TABLE Employees (
    emp_id INT,
    first_name VARCHAR(20),
    last_name VARCHAR(20),
    salary DECIMAL(10,2),
    hire_date DATE,
    dept VARCHAR(20),
    bonus_percent DECIMAL(5,2)
);
```

emp_id	first_name	last_name	salary	hire_date	dept	bonus_percent
1	John	Doe	60000.00	2020-01-15	IT	10.5
2	Jane	Smith	75000.50	2019-03-23	HR	12.0
3	Alice	Green	82000.75	2021-07-10	Sales	15.75
4	Tom	Brown	50000.00	2022-05-01	Marketing	9.0
5	Bob	White	90000.00	2018-11-30	IT	11.25

1. Display the salary of each employee after a 10% increase.
2. Show employee name and annual salary.
3. Display the employee name and half of their salary.
4. Retrieve employees whose salary after a 20% bonus becomes greater than 80000.
5. Find employees whose remaining salary after deducting 10000 is less than 60000.
6. Show the salary after deducting 15% tax from each employee.
7. Display the modulus (remainder) of salary when divided by 10000.
8. Retrieve employees whose salary multiplied by 2 is less than 150000.
9. Show the difference between each employee's salary and the highest salary.
10. Find employees whose salary is equal to the average salary of all employees.